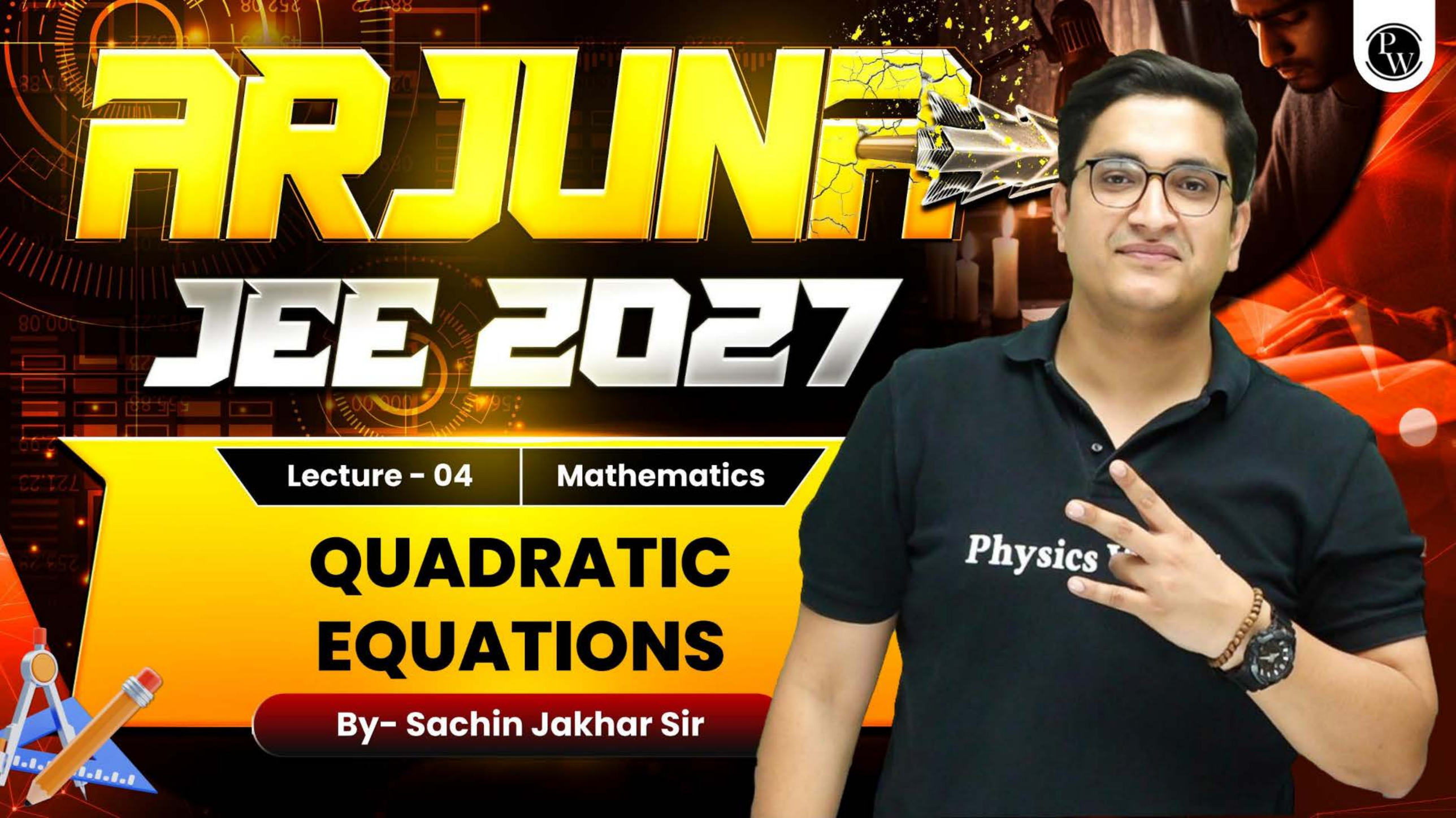
ARJUNA JEE 2027

Lecture - 04 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A Newton's Formula

B Nature of Roots

C Question Practice

D



Question Practice

Question [IIT-JEE-2007, Paper-1]

65%

Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is

A $\frac{2}{9}(p-q)(2q-p)$

B $\frac{2}{9}(q-p)(2p-q)$

C $\frac{2}{9}(q-2p)(2q-p)$

D $\frac{2}{9}(2p-q)(2q-p)$

$x^2 - px + r = 0 \Rightarrow \alpha, \beta$

$x^2 - qx + r = 0 \Rightarrow \frac{\alpha}{2}, 2\beta$

$\alpha + \beta = p$

$\alpha\beta = r$

$\frac{\alpha}{2} + 2\beta = q$

$= \alpha\beta = r$

$\frac{2(2p-q)}{3}, \left(\frac{2q-p}{3}\right)$

$\alpha + \beta = p \rightarrow \alpha = p - \beta = p - \left(\frac{2q-p}{3}\right) = \frac{3p-2q+p}{3}$

$\alpha + 4\beta = 2q$

$-3\beta = p - 2q \Rightarrow \beta = \frac{2q-p}{3}$

$\alpha = \frac{4p-2q}{3}$

Bumper Question

If α & β are roots of equation $x^2 - 4x - 8 = 0$ then find value of following

(i) $\alpha^2 + \beta^2$

(ii) $\left(\frac{\alpha^2}{\beta}\right)^{1/3} + \left(\frac{\beta^2}{\alpha}\right)^{1/3}$

(iii) $\frac{1}{(\alpha-4)^2} + \frac{1}{(\beta-4)^2}$

(iv) $\alpha^4 + \beta^4$

(v) $\alpha^5 + \beta^5$

(vi) $|\alpha^5 - \beta^5|$

(vii) $\frac{\beta}{\alpha^2-8} + \frac{\alpha}{\beta^2-8}$

(viii) $\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)}$

$$\alpha^5+\beta^5 = (\alpha^2+\beta^2)(\alpha^3+\beta^3) - (\beta^2\alpha^3 + \alpha^2\beta^3)$$

(7) $\alpha^2 - 4\alpha - 8 = 0$

$\alpha^2 - 8 = 4\alpha$

$$\frac{\alpha}{\beta^2-8} + \frac{\beta}{\alpha^2-8} = \frac{\beta}{4\alpha} + \frac{\alpha}{4\beta} = \frac{1}{4}\left(\frac{\beta}{\alpha} + \frac{\alpha}{\beta}\right)$$

⑥ $\left|\left(\alpha^5 - \beta^5\right)\right| = \left(\alpha^2 - \beta^2\right)\left(\alpha^3 + \beta^3\right) + \beta^2\alpha^3 - \alpha^2\beta^3$

$= \left|(\alpha-\beta)(\alpha+\beta)(\alpha^3+\beta^3) + \alpha^2\beta^2(\alpha-\beta)\right|$

$= \left|(\alpha-\beta)\right| \cdot \left|\left\{(\alpha+\beta)(\alpha^3+\beta^3) + \alpha^2\beta^2\right\}\right|$

diff. of Roots $= \left|\frac{\sqrt{D}}{a}\right|$

Bumper Question

If α & β are roots of equation $x^2 - 4x - 8 = 0$ then find value of following

(i) $\alpha^2 + \beta^2$

(ii) $\left(\frac{\alpha^2}{\beta}\right)^{1/3} + \left(\frac{\beta^2}{\alpha}\right)^{1/3}$

(iii) $\frac{1}{(\alpha-4)^2} + \frac{1}{(\beta-4)^2}$

(iv) $\alpha^4 + \beta^4$

(v) $\alpha^5 + \beta^5$

(vi) $|\alpha^5 - \beta^5|$

(vii) $\frac{\beta}{\alpha^2-8} + \frac{\alpha}{\beta^2-8}$

(viii) $\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)}$ $= \left(\frac{1}{-3}\right) + \left(\frac{1}{-12}\right) = -\frac{1}{3}\left(1 + \frac{1}{4}\right) = \frac{-5}{12}$

$\alpha^5+\beta^5 = (\alpha^2+\beta^2)(\alpha^3+\beta^3) - (\beta^2\alpha^3 + \alpha^2\beta^3)$

$(\alpha+1)(\beta+1) = \alpha\beta + \beta + \alpha + 1$

(8) $1x^2 - 4x - 8 = 1(x-\alpha)(x-\beta)$ (identity)

$x=2$: $4 - 8 - 8 = (2-\alpha)(2-\beta)$

$-12 = (\alpha-2)(\beta-2)$

$x=-1$: $1 + 4 - 8 = (-1-\alpha)(-1-\beta)$

$-3 = (1+\alpha)(1+\beta)$

Question [JEE (Adv.-2020), Paper-1]

Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$.

Then the value of $ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$.

A. 0

B. 8000

C. 8080

D. 16000

$a^2c - ac^2 + a^2d - ad^2 + b^2c - bc^2 + b^2d - bd^2$

$a^2(c+d) - a(c^2+d^2) + b^2(c+d) - b(c^2+d^2)$

$(c+d)(a^2+b^2) - (c^2+d^2)(a+b)$

72%

$x^2 + 20x - 2020 = 0$ → roots a, b

$a + b = -20$

SBS

$a^2 + b^2 + 2ab = 400$

$ab = -2020$

NEWTON's FORMULA

100% Quesⁿ. आएगा

$ax^2+bx+c=0$ $\begin{cases}\alpha\\ \beta\end{cases}$

$S_n = p\alpha^n \pm q\beta^n$ → $S_1, S_2, S_3, \ldots, S_{100} \ldots$

Hint: $\alpha^n \pm \beta^n$ → #NF

#N.f.:

$$a(S_n) + b(S_{n-1}) + c(S_{n-2}) = 0$$

OR

$$aS_{n+1} + bS_n + cS_{n-1} = 0$$

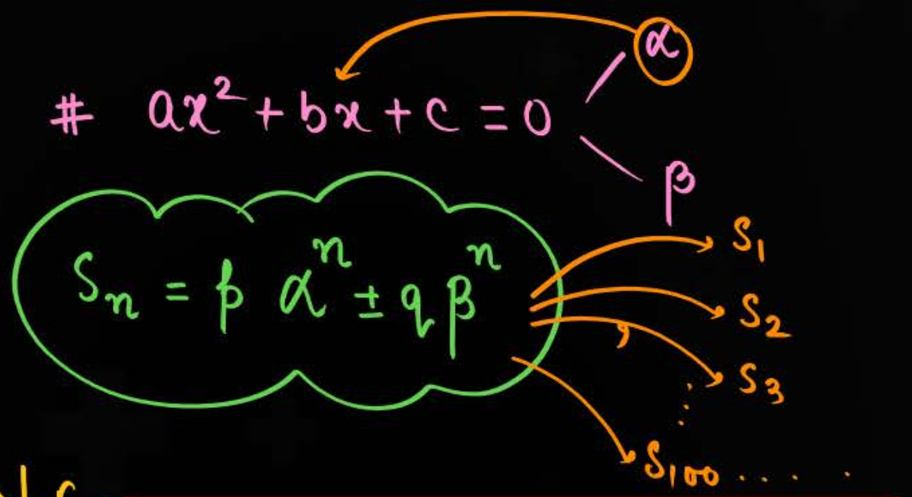
$x^2-5x+6=0$ $\begin{cases}2\\ 3\end{cases}$

$$S_n = 7.2^n - 8.3^n$$

$$1.(S_n) - 5(S_{n-1}) + 6S_{n-2} = 0$$

$n=71$ →

$$S_{71} - 5S_{70} + 6S_{69} = 0$$



N.f. :

$$a(S_n) + b(S_{n-1}) + c(S_{n-2}) = 0$$

$$\text{LHS} = a\left(p\alpha^n \pm q\beta^n\right) + b\left(p\alpha^{n-1} \pm q\beta^{n-1}\right) + c\left(p\alpha^{n-2} \pm q\beta^{n-2}\right)$$

$$= ap\alpha^n + bp\alpha^{n-1} + cp\alpha^{n-2} \pm \left(aq\beta^n + bq\beta^{n-1} + cq\beta^{n-2}\right)$$

$$= p\alpha^{n-2}\left\{a\alpha^2 + b\alpha + c\right\} \pm q\beta^{n-2}\left\{a\beta^2 + b\beta + c\right\}$$

$$= 0$$

$$= \text{RHS}.$$

RAPID FIRE ROUND

1. Roots of Quadratic $x^2 - 5x - 7 = 0$ are α & β and $S_n = \alpha^n + \beta^n$ then $\frac{S_{10} - 7S_8}{S_9} =$ _____ .

5

NF: $1(S_n) - 5(S_{n-1}) - 7(S_{n-2}) = 0$

2. Roots of Quadratic $3x^2 - x - 11 = 0$ are p & q and $a_n = 7p^n - 6q^n$ then $\frac{a_{20} + 11a_{19}}{a_{21}} =$ _____ .

3

NF

$3a_n - a_{n-1} - 11a_{n-2} = 0$

$n = 21$

$3a_{21} - a_{20} - 11a_{19} = 0$

$3a_{21} = a_{20} + 11a_{19}$

$n = 10$

$S_{10} - 5S_9 - 7S_8 = 0$

$\frac{S_{10} - 7S_8}{S_9} = 5$

Question [JEE Mains-2021

Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$;

Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then the value of p_n^2 is

$x^2 - (1)x + (-1) = 0$ ⟨ α, β

$x^2 - x - 1 = 0$ ⟨ α, β.

$p_{n+1} - p_n - p_{n-1} = 0$.

$29 - 11 = p_n$

$18 = p_n$

$p_n^2 = 18^2$

$= 324$ Ans.

Note :

$3x^2 + 11 = 0 \begin{cases} \alpha \\ \beta \end{cases}$

$S_n = \alpha^n + \beta^n$

S-I : $3S_n + 11S_{n-1} = 0$ ✗

S-II : $3x^2 + 0x + 11 = 0$

$\rightarrow 3S_n + 0S_{n-1} + 11S_{n-2} = 0$ ✓

Question [JEE Mains-2025]

68

Let α and β be the roots of $x^2 + \sqrt{3}x - 16 = 0$, and γ and δ be the roots of $x^2 + 3x - 1 = 0$. If $P_n = \alpha^n + \beta^n$ and $Q_n = \gamma^n + \delta^n$, then $\frac{P_{25} + \sqrt{3}P_{24}}{2P_{23}} + \frac{Q_{25} - Q_{23}}{Q_{24}}$ is equal to

A. 3

B. 4

C. 5

D. 7

$x^2 + \sqrt{3}x - 16 = 0 \Rightarrow \alpha, \beta$

$P_n = \alpha^n + \beta^n$

NF: $P_n + \sqrt{3}P_{n-1} - 16P_{n-2} = 0$

$n = 25$

$P_{25} + \sqrt{3}P_{24} = 2 \times 8 P_{23}$

$x^2 + 3x - 1 = 0 \Rightarrow \gamma, \delta$

$Q_n = \gamma^n + \delta^n$

NF: $Q_n + 3Q_{n-1} - Q_{n-2} = 0$

$n = 25$

$Q_{25} + 3Q_{24} - Q_{23} = 0$

$\frac{Q_{25} - Q_{23}}{Q_{24}} = -3$

$8 + (-3)$

Question [JEE Mains-2021]

If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ each positive integer n, then the value of $\left(\frac{P_{17}P_{20}+5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19}+5\sqrt{2}P_{18}^2}\right)$ is equal to _____.

HW.
(1)

Question [JEE Mains-2022]

For a natural number n, let $\alpha_n = 19^n - 12^n$. Then, the value of $\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$ is _____.

Quad.

Roots → 19, 12.

$x^2 - (19+12)x + 19\times12 = 0$ (19, 12)

NF

$\alpha_n - 31\alpha_{n-1} + 19\times12\alpha_{n-2} = 0$

$n = 10$

$\alpha_{10} - 31\alpha_9 + 19\times12\alpha_8 = 0$.

$\alpha_n = 19^n - 12^n$

(4) $= \frac{19\times3\times4}{57} = \frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$.

Ans.

Question [JEE Mains-2020]

Let α and β be the roots of the equation $x^2 - x - 1 = 0$.

If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which of the following statements is not true?

A (F) $p_5 = p_2 \cdot p_3$

B (T) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

C (T) $p_3 = p_5 - p_4$

D (T) $p_5 = 11$

$\alpha\beta = -1$

α, β

$p_2 = \alpha^2 + \beta^2$

$= (\alpha+\beta)^2 - 2\alpha\beta$

$p_2 = 1 - 2(-1) = 3$

$p_3 = \alpha^3 + \beta^3$

$= (\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta)$

$= 1 \times (3 - (-1))$

$p_3 = 4$.

$p_4 = \alpha^4 + \beta^4$

$k=2$

$k=5$

$\# \; p_k = \alpha^k + \beta^k \xrightarrow{k=1} p_1 = \alpha' + \beta' = \text{Sum of Roots}$

NF $\quad p_k - p_{k-1} - p_{k-2} = 0$

$p_5 - p_4 - p_3 = 0$.

$p_1 = \frac{-(-1)}{1}$

$\# \; p_1 = 1$

$\# \; p_2 = 3$

$P_1 = \alpha' + \beta' = 1$

$P_2 = \alpha^2 + \beta^2 = 3$ ✓

$P_3 = \alpha^3 + \beta^3 = 4$ ✓

$P_4 = \alpha^4 + \beta^4 = 7$.

$P_5 = \alpha^5 + \beta^5 = 11$

$P_6 = \alpha^6 + \beta^6 = 18$

⋮

NF

$$P_k - P_{k-1} - P_{k-2} = 0$$

$k = 4$

$$P_4 - P_3 - P_2 = 0$$

$$P_4 = P_3 + P_2$$

$k = 5$

$$P_5 - P_4 - P_3 = 0$$

$$P_5 = P_4 + P_3$$

Question [JEE Mains-2025]

Reverse

Let $P_n = \alpha^n + \beta^n, n \in N$. If $P_{10} = 123, P_9 = 76, P_8 = 47$ and $P_1 = 1$, then the quadratic equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is:

A $x^2 - x + 1 = 0$

B $x^2 + x - 1 = 0$

C $x^2 - x - 1 = 0$

D $x^2 + x + 1 = 0$

$\# \quad P_{10} = P_9 + P_8$

$P_{10} - P_9 - P_8 = 0$

$n = 10$

$P_n - P_{n-1} - P_{n-2} = 0$

$x^2 - x - 1 = 0$

$x^2 - x - 1 = 0 \Big/ \begin{matrix} \alpha \\ \beta \end{matrix}$

$\alpha + \beta = 1, \quad \alpha\beta = -1$

$\frac{\alpha+\beta}{\alpha\beta} = -1$

$x^2 - \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x + \frac{1}{\alpha\beta} = 0$

$x^2 + x - 1 = 0 \Big/ \begin{matrix} \frac{1}{\alpha} \\ \frac{1}{\beta} \end{matrix}$

Question [JEE Mains-2023]

If a and b are the roots of equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ is equal to

Question [JEE Mains-2023]

Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to

HW

A $-64\sqrt{2}$

B $-128\sqrt{2}$

C -64

D -128

Question [JEE Mains-2023]

Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$ is equal to

A 729

B 72

C 81

D 9

Question [JEE Mains-2024]

Let α, β be the distinct roots of the equation $x^2 - (t^2 - 5t + 6)x + 1 = 0, t \in R$ and $a_n = \alpha^n + \beta^n$. Then the minimum value of $\frac{a_{2023} + a_{2025}}{a_{2024}}$ is

A $\frac{1}{4}$

B $-\frac{1}{2}$

C $-\frac{1}{4}$

D $\frac{1}{2}$

Min. value of $(t^2 - 5t + 6) = -\frac{1}{4}$.

$\frac{dy}{dt} = 2t - 5 = 0$

$t = \frac{5}{2}$

$t^2 - 5t + 6\Big|_{\text{min.}} = \left(\frac{5}{2}\right)^2 - 5 \times \frac{5}{2} + 6$

$= -\frac{1}{4}$.

NF

$a_n - (t^2 - 5t + 6)a_{n-1} + a_{n-2} = 0$

$n = 2025$.

$\frac{a_{2025} + a_{2023}}{a_{2024}} = (t^2 - 5t + 6)$



NATURE OF ROOTS

#CHAUDHARY kaun hai??

$D = b^2 - 4ac$

$(a, b, c \in R,\ a \neq 0) \Rightarrow ax^2 + bx + c = 0$ ⟨ α, β.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$D = b^2 - 4ac$

$D > 0 \Rightarrow$ Roots are Real & Distinct (different)

$D = 0 \Rightarrow$ Roots are Real & equal
OR
Quad. becomes Perfect Square

$D < 0 \Rightarrow$ Roots are non-Real
OR
Roots are Imaginary

$ax^2 + bx + c = 0$

$\mathcal{D} = b^2 - 4ac = 0$

$\# \quad b^2 = 4ac$ ✓

$\frac{b^2}{4a} = c$

$ax^2 + bx + \frac{b^2}{4a} = 0$

$4a^2x^2 + 4abx + b^2 = 0$

$(2ax + b)^2 = 0$

DO IT BY YOURSELF (DIBY)

DIBY-04

If α and β are roots of the equation $x^2 + px + \frac{3p}{4} = 0$, such that $|\alpha - \beta| = \sqrt{10}$, then p belongs to the set

(A) $\{2, -5\}$ (B) $\{-3, 2\}$ (C) $\{3, -5\}$ (D) $\{-2, 5\}$

[Ans. : (D)]

DIBY-05

If α, β are the roots of $ax^2 + bx + c = 0$ and $c \neq 0$ then the value of $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$ in terms of a, b, c is

(A) $\frac{b^2-2ac}{ac}$ (B) $\frac{2ac-b^2}{ac}$ (C) $\frac{b^2-2ac}{a^2c^2}$ (D) $\frac{b^2}{2}$

[Ans. : (C)]

DIBY-06

$x^2-5x+6 + x^2-2x-3 + x^2-x-2 = 0 \longrightarrow 3x^2-8x+1 = 0$ $\begin{matrix}\alpha \\ \beta\end{matrix}$

Let α and β be the roots of the quadratic equation $(x-2)(x-3) + (x-3)(x+1) + (x+1)(x-2) = 0$. Find the value of

$$\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)} + \frac{1}{(\alpha-3)(\beta-3)}$$

[Ans. : 0]

$(\alpha+\delta)-(\beta+\delta)=(\alpha-\beta)$

DIBY-07

If α, β are the roots of $ax^2+bx+c=0, (a\neq 0)$ and $\alpha+\delta, \beta+\delta$ are the roots of $Ax^2+Bx+C=0, (A\neq 0)$ for some constant δ, then prove that, $\frac{b^2-4ac}{a^2}=\frac{B^2-4AC}{A^2}$.

[JEE-2000]

$\frac{\sqrt{D}}{|a|}=\frac{\sqrt{D}}{|A|}$

DIBY-08

If α and β be two roots of the equation $x^2-64x+256=0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}}+\left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is

$\frac{\alpha^{3/8}}{\beta^{5/8}}+\frac{\beta^{3/8}}{\alpha^{5/8}}$

$\frac{b^2-4ac}{a^2}=\frac{B^2-4AC}{A^2}$

(A) 1 (B) 3 (C) 4 (D) 2

$\frac{\alpha'+\beta'}{(\alpha\beta)^{5/8}}$

[Ans. : (D)]

DIBY-09

Let α, β be the roots of the equation $x^2+2\sqrt{2}x-1=0$. The quadratic equation, whose roots are $\alpha^4+\beta^4$ and $\frac{1}{10}(\alpha^6+\beta^6)$, is:

[JEE Mains-2024]

(A) $x^2-190x+9466=0$ (B) $x^2-195x+9466=0$

(C) $x^2-195x+9506=0$ (D) $x^2-180x+9506=0$

[Ans. : (C)]

DIBY-10

If the equation $a(b-c)x^2+b(c-a)x+c(a-b)=0$ has equal roots, where $a+c=15$ and $b=\frac{36}{5}$, then a^2+c^2 is equal to [JEE Mains-2025]

[Ans. : 117]

DIBY-11

Let $\alpha \in \mathbb{R}$ and let α, β be the roots of the equation $x^2+60\sqrt[4]{x}+a=0$.
If $\alpha^4+\beta^4=-30$, then the product of all possible values of a is [JEE Mains-2023]

[Ans. : 45]

DIBY-12

If 2 and 6 are the roots of the equation $ax^2+bx+1=0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is : [JEE Mains-2024]

(A) $x^2+8x+12=0$
(B) $2x^2+11x+12=0$
(C) $4x^2+14x+12=0$
(D) $x^2+10x+16=0$

[Ans. : (A)]

"Newton's formula"

DIBY-13

Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n, n = 1, 2, 3 \ldots$., then

[JEE Mains-2020]

(A) $6S_6 + 5S_5 = 2S_4$

(B) $6S_6 + 5S_5 + 2S_4 = 0$

(C) $5S_6 + 6S_5 = 2S_4$

(D) $6S_6 + 6S_5 + 2S_4 = 0$

[Ans. : (C)]

DIBY-14

If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then

[JEE Mains-2024]

(A) $2S_{12} = S_{11} + S_{10}$

(B) $S_{12} = S_{11} + S_{10}$

(C) $2S_{11} = S_{12} + S_{10}$

(D) $S_{11} = S_{10} + S_{12}$

[Ans. : (B)]

DIBY-15

Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to

[JEE Mains-2024]

[Ans. : 4]

DIBY-16

Let $\alpha, \beta; \alpha > \beta$, be the roots of the equation $x^2 - \sqrt{2}x - \sqrt{3} = 0$. Let $P_n = \alpha^n - \beta^n, n \in N$. Then $(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$ is equal to:

[JEE Mains-2024]

(A) $10\sqrt{2}P_9$ (B) $10\sqrt{3}P_9$ (C) $11\sqrt{2}P_9$ (D) $11\sqrt{3}P_9$

[Ans. : (B)]

DIBY-17

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_e = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is:

[JEE Mains-2021]

(A) 2 (B) 1 (c) 4 (d) 3

[Ans. : (A)]

DIBY-18

Let $\alpha, \beta(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in N$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to.

[JEE Mains-2022]

[Ans. : 16]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Basic Maths (4 Sittings Each of 1 Hr.)

Exercise-03(BRIDGE): COMPLETE

Exercise-04(JEE ADV. LEVEL):COMPLETE

It is never about end result

It's always about Journey

Thank You

#futureIITians